LegalSimple: An Explainable AI Framework for Clause Extraction, Summarization, and Risk Flagging in Legal Documents

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***Abstract*—Legal documents, such as contracts, policies, and regulatory notices, can be long, superficially organized, and contain a lot of technical language—a barrier for non-lawyers. We introduce the LegalSimple framework to reduce the complexity of document comprehension while maintaining factual accuracy and document traceability. LegalSimple integrates semantic retrieval based on FAISS with a retrieval-augmented generation (RAG) module to develop clause-level explanations, bilingual summaries, and risk-based indicators. Each clause is assigned to a Red, Yellow, or Green risk category to indicate possible risk, which helps users prioritize obligation or liability in compliance with a "traffic-light" style of contract analysis. A reinforcement learning (RL) module supports the adjustment of each risk threshold and improves trust in system results over time and user feedback. LegalSimple's modules not only support clause extraction and plain-language but also support extractive summarization in both English and Hindi. Together, the components of the system promote the use of transparent and actionable AI support for lawyers and non-legal stakeholders.**

***Index Terms*—Legal AI, Explainable AI, Retrieval-Augmented Generation, FAISS, Risk Analysis, Clause Extraction, Summarization, Reinforcement Learning**

I. Introduction

Legal systems, regardless of jurisdiction, depend on documents—contracts, government regulations, and compliance policies—to delineate rights and obligations. While such documents precisely establish rights and obligations, they tend to be written in technical legal language, limiting the public's understanding. Thus, individuals and organizations often make mistakes in identifying their duties, liabilities, and the risks they may face.

Large Language Models (LLMs) have shown remarkable fluency in text generation, but their integration into legal practice is still limited by challenges of factual accuracy, explainability, and accountability. Although several summarization or information-retrieval tools exist, they frequently address the general context of a document rather than clause-level detail, often resulting in incomplete or misleading outputs when the stakes are high.

Recent work on Retrieval-Augmented Generation (RAG) has demonstrated that augmenting LLMs with structured retrieval mechanisms can drastically improve factual grounding. However, most existing RAG implementations are not specifically designed for legal documents and often fail to recognize clause boundaries, risk implications, and multilingual capabilities.

In light of these challenges, we introduce **LegalSimple**, an AI tool designed for legal document analysis, to address these problems. The following are this paper's primary contributions:

- **Semantic Retrieval:** Quick clause-level retrieval with FAISS.
- **Risk Categorization:** Reinforcement Learning (RL)-based border threshold optimization is used to automatically classify retrieved sentences into three risk categories (Red, Yellow, and Green).
- **Clause Extraction and Plain Language Transformation:** This process identifies duties, fines, and clauses pertaining to compliance and converts them into plain language.
- **Two-Language Summarization:** To promote multilingual accessibility, brief, readable summaries of clauses are created in two languages.

II. Related Work

The agility of research in areas like contract analytics, judgment prediction, and legal explainability within the intersection of artificial intelligence and law has accelerated dramatically. While LEDGAR included detailed annotations for clauses that allowed for classified legal texts, earlier datasets like CUAD created baselines for contracts through standardized evaluation and clause annotation. These developments have played a key role in improving legal usability within NLP research studies.

In the context of such legal corpora to pretrained AI model like Legal BERT, authority by transformer architectures in particular illustrates that transformer models tailored for law similarity prediction are better suited than their general counterparts including BERT The InLegalBERT model took this concept and applied it to India by incorporating local legal language and multilingual writing styles.

At the same time, the Retrieval-Augmented Generation (RAG) framework offered an effective method to merge gen-

erative reasoning with factual retrieval. In legal tasks, this approach helps cut down on errors and improves how evidence can be traced. Research on predicting legal judgments has redefined decision prediction as a reading comprehension challenge. This shift has strengthened the connection between model outputs and human understanding.

Explainability has also become a major focus in legal AI. Benchmarks like ERASER set up metrics for rationale extraction and highlighted the need for clear model outputs. On the other hand, studies such as **"Attention Is Not Explanation"** cautioned against reading too much into attention weights as trustworthy signs of interpretability.

In the same direction, RAG continues to serve as a bridge between reasoning and factual retrieval. It helps minimize hallucination and improves traceability. Research efforts predicting legal judgments have also treated judgments as comprehension-based tasks. This approach enhances reasoning consistency in predictions. Explainability remains central to this work. ERASER and similar benchmarks emphasize the importance of rationales in clear outputs. However, cautionary studies like **"Attention Is Not Explanation"** highlight the limitations of depending only on attention mechanisms for understanding.

III. Positioning

While datasets such as CUAD and LEDGAR are valuable for studying contracts, they are primarily developed to capture Western legal systems and the English language. As a consequence the regional divisions, particularly in Indian Documents and Legal texts are largely forgotten. There was some successful application of Legal-BERT, however it does not completely consider multilingual clauses and cultural characteristics of legal drafting. InLegalBERT offers some relief to this by pre-training over Indian legal text but its utility in retrieval-generation scenarios, for risk assessment generation, has not been explored due to the extravagant pre-training cost. On the other hand, Retrieval-Augmented Generation (RAG) approaches made a progress in factual grounding but they are weak at clause-level details and no explanation can be obtained from them to provide more understanding. Likewise, explainability benchmarks such as ERASER center around textual rationales and lack to incorporate reinforcement learning for more adaptive and interpretable decision making.

LegalSimple is positioned at the intersection of these evolutionary forces, bringing their strengths together to form an integrated and practical framework. The system integrates:

- OCR based pre-processing and normalization of PDFs and scanned contracts,
- Domain-specific semantic retrieval using fine-tuned InLegal- BERT embeddings,
- Bilingual output with RAG-based clause-aware language model summaries, and simplifications
- An RL-guided risk-flagging module which provides rationale data, covers spans and alternatively adjusts the level of decision certainty user feedback.

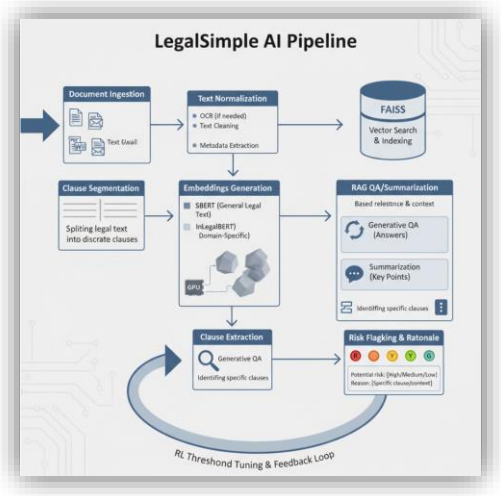
This comprehensive approach is intended to result in a legal sub-standing system that is domain-aware, interpretable and adaptive.

IV. Research Problem and Goals

Further, legal documents (e.g., contracts, policies and notices etc) are long in size, lack of a uniform structure or format [1], [2]. While pre-trained languagemodels such as Legal-BERT [3] and InLegalBERT [4] have advanced legal NLP, existing methods still face critical challenges:

- **Relevance at clause level:** Advanced language models are not able to identify and retrieve sub-clause level clauses containing legally relevant information.
- **Risk explanation:** Existing retrieval and summarypipelines do not generate actionable risk predictions with interpretable justification
- **Explainability:** the reasoning behind an automated decision cannot always be backed up with evidence ESP According to some potentially high-stakes legal decisions, reduce trust [6], [7].
- **Local adaptation:** Indian legal texts show specific terminology, multilingual text and statutory conventions that pretrained models like those trained with Western datasets may not be able to capture well [4].

Addressing these gaps needs a system which have to integrates robust text extraction, semantic retrieval, risk-aware reasoning, explainable outputs, and localization for Indian legal documents.



A. Objectives

The main aims of this study are:

- **Clause-level Semantic Retrieval:** Design a FAISS-based embedding index for fast retrieval of legally relevant clauses from contracts and policies [1], [2], [5].
- **Risk Classification and Explainability:** Develop a risk stratification model using Red–Yellow–Green scheme for classifying radiological exams based on reinforcement learn-

ing, in which explanation spans are underlined to facilitate explainable reasoning process [8], [9].

- **Clause Extraction and Simplification:** System-off automatically extract clauses and simple obligations, penalties, or com pliance-relevant spans; rephrase to understandable language [**?**], [6].
- **Cross-lingual Summarisation:** Generate short summaries in many languages to improve access and usability in multilingual legal context [4].
- **Domestic Fine-tuning:** Fine-tune pre-trained legal language models (InLegalBERT) with Indian statutory and contractual corpora for domain or regional performance [4].
- **End-to-End Workflow Integration:** Create a pipeline that interconnects text extraction, clause segmentation, semantic retrieval and risk assessment to deliver insights for use in making decisions.

By achieving these targets, we hope that LegalSimple can provide a transparent, accessible and actionable technique for encoding and assessing legal documents on top of the trade- off between accurateness, interpretability, as well usability towards different types of users.

V. System Architecture / Methodology

A. Data Collection and Preprocessing

Legal documents such as contracts, policies, and judgments are typically unstructured and lengthy. The preprocessing stage ensures that the text is standardized and legally meaningful before feeding it into downstream models.

- **Optical Character Recognition (OCR):** Scanned documents and PDFs are converted to machine-readable text using *Tesseract OCR*.
- **Text Cleaning:** Unnecessary metadata such as page numbers, signatures, and headers are removed using regular expressions and rule-based filters.
- **Clause Segmentation:** Legal documents are segmented into clauses using *spaCy dependency parsing* and handcrafted linguistic patterns (keywords such as *"Whereas," "Provided that,"* and *"Hereinafter"*). This enables clause-level semantic understanding and retrieval.
- **Normalization:** Text is converted to lowercase, punctuation and spacing are standardized, and stop-word removal is performed selectively to preserve legal significance.

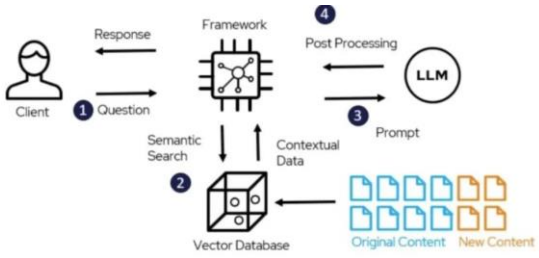
B. Semantic Embedding Generation using InLegalBERT

The preprocessed clauses are transformed into dense semantic vectors using **InLegalBERT**, a transformer model fine-tuned for Indian legal texts. InLegalBERT extends **Legal-BERT**, originally trained on U.S. case law and statutes, by further fine-tuning it on **CUAD (Contract Understanding Atticus Dataset)** and additional Indian legal corpora.

- **Base Model:** Legal-BERT (based on BERT-Base-Uncased, trained on U.S. legal corpora).
- **Intermediate Fine-Tuning:** InLegalBERT trained on CUAD to enhance contextual understanding of contractual terms.
- **Vectorization:** Clause-level sentence embeddings extracted from the [CLS] token.
- **Sentence Transformer Layer:** Sentence-BERT layer is integrated to refine embedding similarity scoring for long clauses.
- **Indexing:** All embeddings are indexed using **FAISS (Facebook AI Similarity Search)** for efficient vector retrieval.

C. Retrieval-Augmented Generation (RAG) with LangChain

- To enable clause-aware summarization and question answering, LegalSimple adopts a **RAG (Retrieval-Augmented Generation)** approach integrated with **LangChain**.
- **Retriever:** FAISS retrieves the top-k most semantically relevant clauses based on query embeddings.
- **Generator:** The retrieved context is passed to a text-generation model to produce clause-wise explanations and summaries.
- **LangChain Integration:** LangChain manages the data flow between retriever and generator, enabling pipeline modularity, multi-language prompts, and tool chaining.



D. Reinforcement Learning–Based Risk Flagging

- **Input Representation:** Clause embeddings generated by **InLegalBERT** serve as the input state s_t.Each state represents a clause's contextual semantics derived from the retrieval stage.
- **RL Policy Network:** The agent is trained using **Proximal Policy Optimization (PPO)**, a stable policy-gradient algorithm. It predicts an action

$a_t \in Red, Yellow, Green$ for each clause. Policy parameters are updated iteratively based on expert feedback.

- **Risk Classification Scheme: High Risk (Red):** Ambiguous or high-liability clauses. **Moderate Risk (Yellow):** Unclear or partially compliant clauses. **Low Risk (Green):** Clear, standard, or low-liability clauses.

Reward Function Summary

Case	Meaning	Reward
Correct High-Risk Detection	Accurate identification of critical clause	+2
Correct Medium/Low Classification	Correct safe or moderate clause flag	+1
	Slight error (e.g. high → medium)	−3
Minor Misclassification	Severe error (e.g. high → low)	−5

E. Bilingual Summarization and Simplification

- **Summarization Model:** The summarization engine is based on **MILD-SUM**, an Indian Legal Dataset developed for bilingual summarization tasks.
- **Simplification Model:** The generative model used is **T5-MILD-SUM**, fine-tuned on bilingual data to produce clause-aware, plain-language summaries.
- **Translation Module:** *IndicTrans2* is employed for translation consistency between English and Hindi outputs.

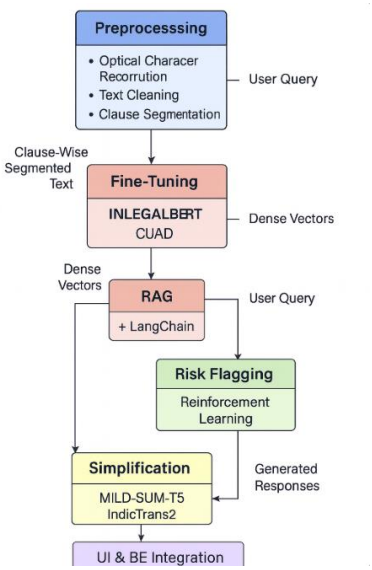
G. Fine-Tuning Strategy (CUAD → InLegalBERT → MILD-SUM)

- **CUAD Pretraining:** Legal-BERT is first fine-tuned on CUAD, which focuses on contract risk and clause identification.
- **InLegalBERT Domain Adaptation:** The CUAD-fine-tuned model is further adapted on Indian legal corpora for jurisdiction-specific semantics.
- **MILD-SUM Task-Specific Fine-Tuning:** Finally, the model is fine-tuned on the MILD-SUM dataset to inculcate Indian Legal Knowledge and bilingual summarization and clause simplification.

F. System Integration

- **Front-End:** Web interface allowing users to upload contracts and query clauses interactively.
- **Back-End:** LangChain orchestrates preprocessing, embedding retrieval, RAG, and summarization modules.
- **APIs:** RESTful endpoints for document ingestion, retrieval, risk flagging, and summary generation.
- **Storage:** Vector and metadata storage managed through FAISS

LEGALSIMPLE Pipeline Structure



VI. Results & Analysis

A. Model Development Status

LegalSimple is currently **Under Active Development**. The following results are **partial yet indicative**, as fine-tuning, reinforcement calibration, and bilingual evaluation are ongoing. All presented figures highlight comparative and component-level outcomes from **CUAD-based fine-tuning**, **Learning (RL)–based risk flagging**.

B. Performance Improvements through Fine-Tuning

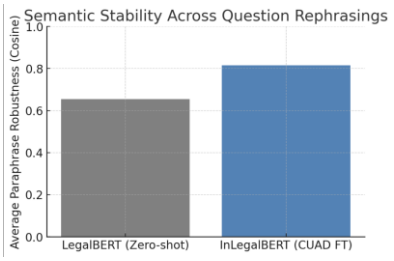


Figure 1 Semantic Stability Across Question Rephrasings

This figure compares the *semantic robustness* between LegalBERT (zero-shot) and InLegalBERT (CUAD fine-tuned) using cosine similarity of paraphrase embeddings. The fine-tuned InLegalBERT achieves ≈ 0.81 average robustness (vs 0.66 baseline), showing higher contextual stability across reworded queries.

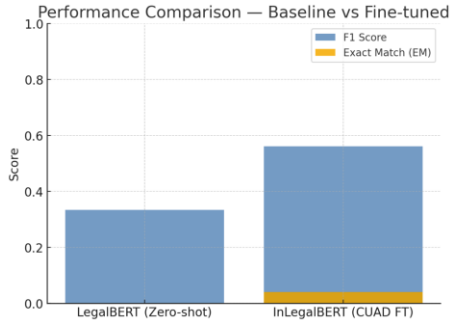


Figure 2. Performance Comparison Baseline vs Fine-tuned

Here, F1 and Exact-Match (EM) scores demonstrate the benefit of CUAD fine-tuning.

- *LegalBERT (Zero-shot):* F1 ≈ 0.33 | EM = 0.00
- *InLegalBERT (CUAD FT):* F1 ≈ 0.56 | EM = 0.00
  This 23-point F1 improvement validates effective legal-domain adaptation.



Figure 3. Clause-wise Performance Comparison

Clause-level F1 analysis across 20 contract categories reveals that **InLegalBERT (Fine-tuned)** consistently outperforms **LegalBERT (Zero-shot)**, particularly in complex clauses such as *Indemnification*, *Liability*, and *Force Majeure*. But still requires improvement as this is just a smoke test not final evaluation since the model is yet not completely ready.

C. Boundary Trimming Enhancement

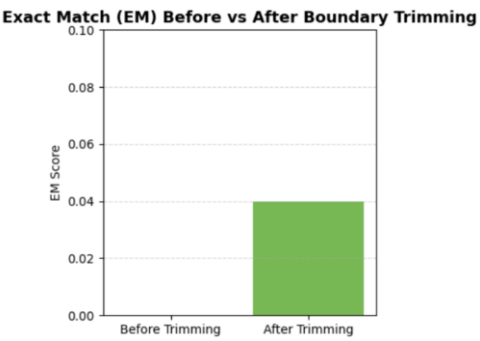


Figure 4. Exact Match (EM)

Here, Post-processing through boundary trimming improved exact-span accuracy: **EM ↑ from 0.00 → 0.04**, indicating reduced span misalignment in extracted clause predictions.

D. Semantic and Retrieval Evaluation

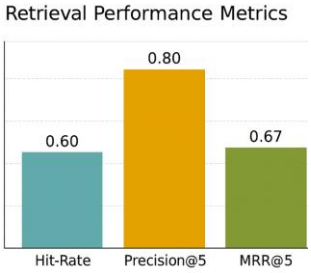


Fig 5. Semantic and Retrieval Evaluation

This retrieval result reflects the system's early performance right after fine-tuning InLegalBERT on the CUAD dataset. The metrics — Hit-Rate = 0.60, Precision@5 = 0.80, and MRR@5 = 0.67 — indicate that the model retrieves relevant legal clauses in most queries, with correct answers often ranking within the top 2–3 results. These values show that CUAD fine-tuning significantly improved semantic retrieval accuracy, even before any additional fine-tuning or reinforcement steps were applied**.**

E. Reinforcement Learning – Based Risk Flagging

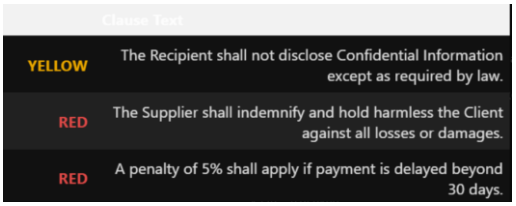


Fig 6. RL Flagging

This visual showcases clause-wise risk interpretation, where each color reflects the RL agent's decision under the reward scheme on a sample legal document.

Quantitative RL Classifier Metrics

Accuracy	0.91
Macro F1	0.8113
Yellow F1	0.9283
Red F1	0.8968
Green F1	0.6087

These preliminary scores confirm strong separation between risk categories while preserving interpretability.

F. Threshold Optimization for Risk Flagging Stability

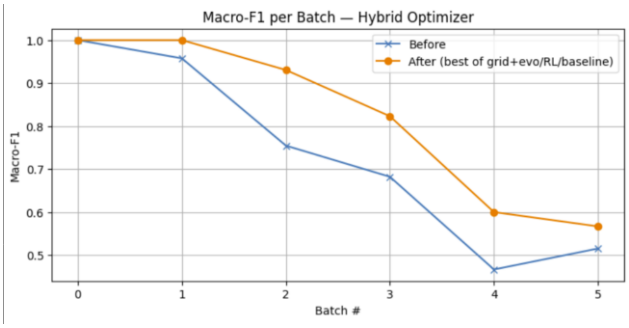


Fig 7. Optimizing RL

To ensure balanced classification across all risk levels, we implemented a **Hybrid Threshold Optimizer** combining coarse-to-fine grid search with evolutionary refinement. This method adaptively tuned the clause classification thresholds (tR, tY, τ) to maximize Macro-F1 on validation batches. The optimizer ensures risk boundaries remain logically ordered (Red < Yellow < Green) while improving per-batch performance stability.

VI. Conclusion & Future Work

The **LegalSimple** project is currently in its developmental stage, showing promising results in clause-level understanding, retrieval, and risk flagging using **InLegalBERT fine-tuned on CUAD**. The system achieved strong preliminary outcomes, including a **Macro-F1 of 0.81** and **91% accuracy** in risk classification, alongside effective retrieval metrics. However, this work is **still ongoing** — further **fine-tuning on the Indian MILD-SUM dataset**, along with the **bilingual summarization and simplification modules**, is yet to be completed. Once these components are integrated, LegalSimple is expected to evolve into a complete, explainable, and bilingual legal assistant capable of simplifying complex legal language for both experts and non-experts.

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